

From Associations to Activations

Comparing Behavioral and Hidden-State Semantic Geometry in LLMs

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Motivation

A classic cognitive-science question (Baker, Saxe, and Tenenbaum, 2009): If you only observe a system's **behavior**, how much can you recover about the structure of its internal **cognitive activations**?

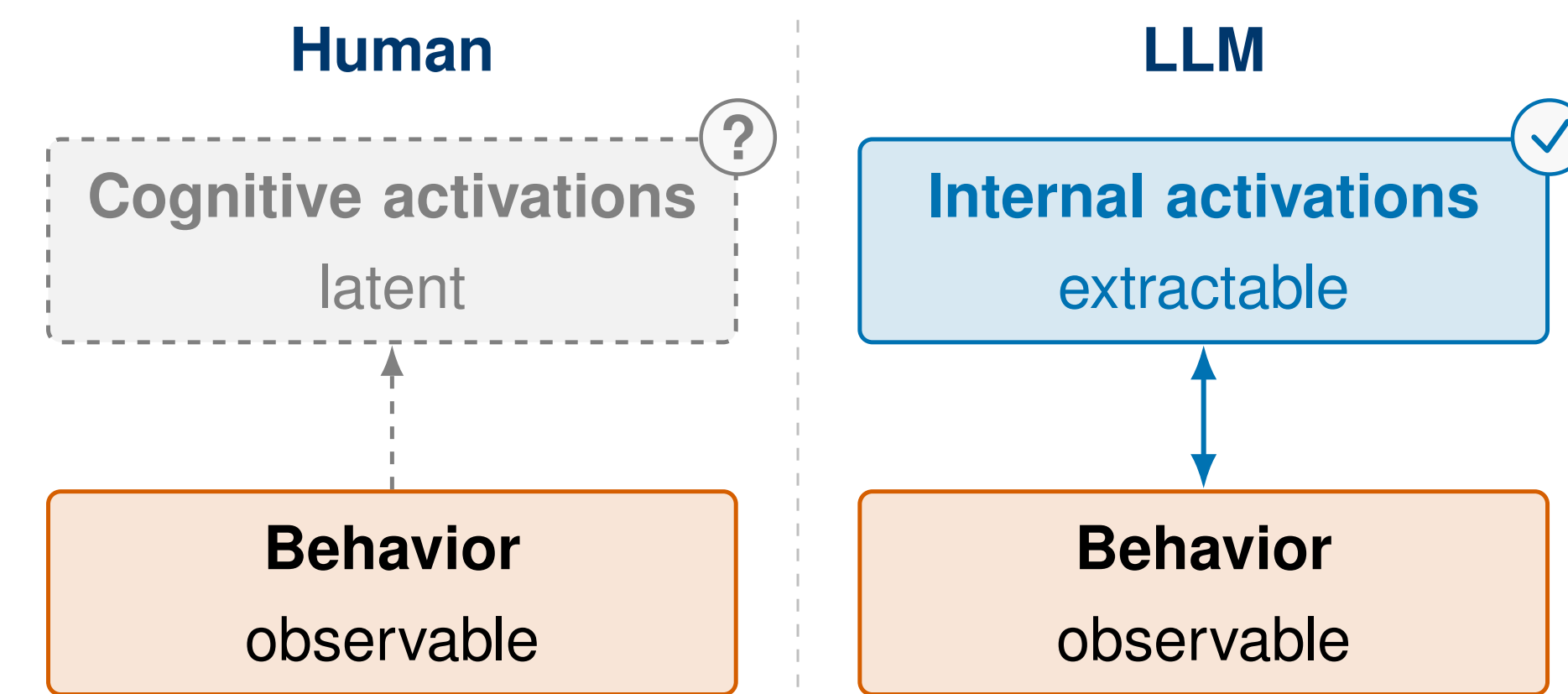


Figure 1. In LLMs, internal activations are extractable, so the behavior→internals inference is testable.

Comparing Behavior and Internals

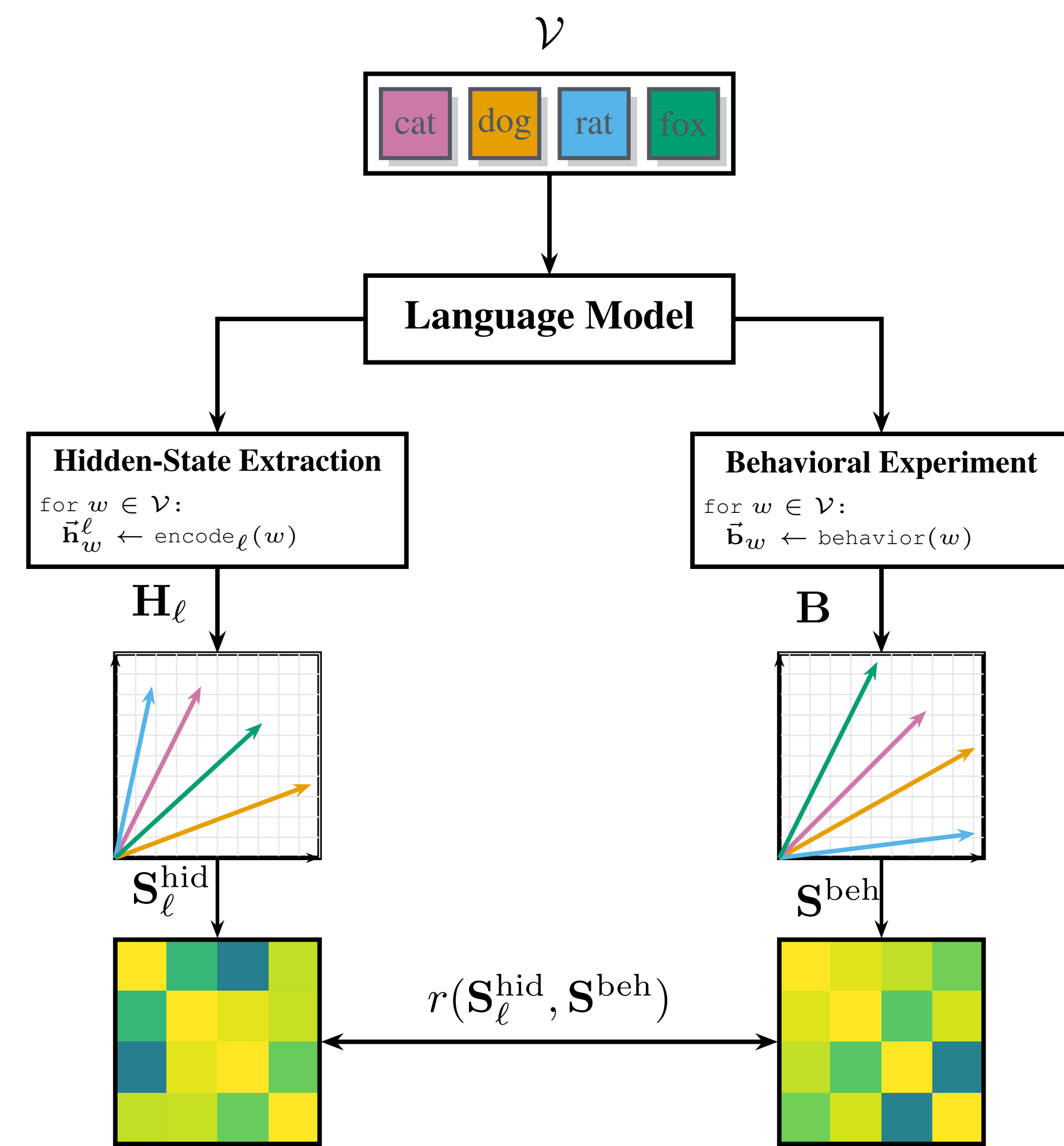


Figure 2. Internal path (left): extract layerwise hidden states $\rightarrow S_{\ell}^{\text{hid}}$. Behavioral path (right): run behavioral tasks $\rightarrow S_{\ell}^{\text{beh}}$. RSA correlates the two: $r(S_{\ell}^{\text{hid}}, S_{\ell}^{\text{beh}})$.

Behavioral Semantic Geometry

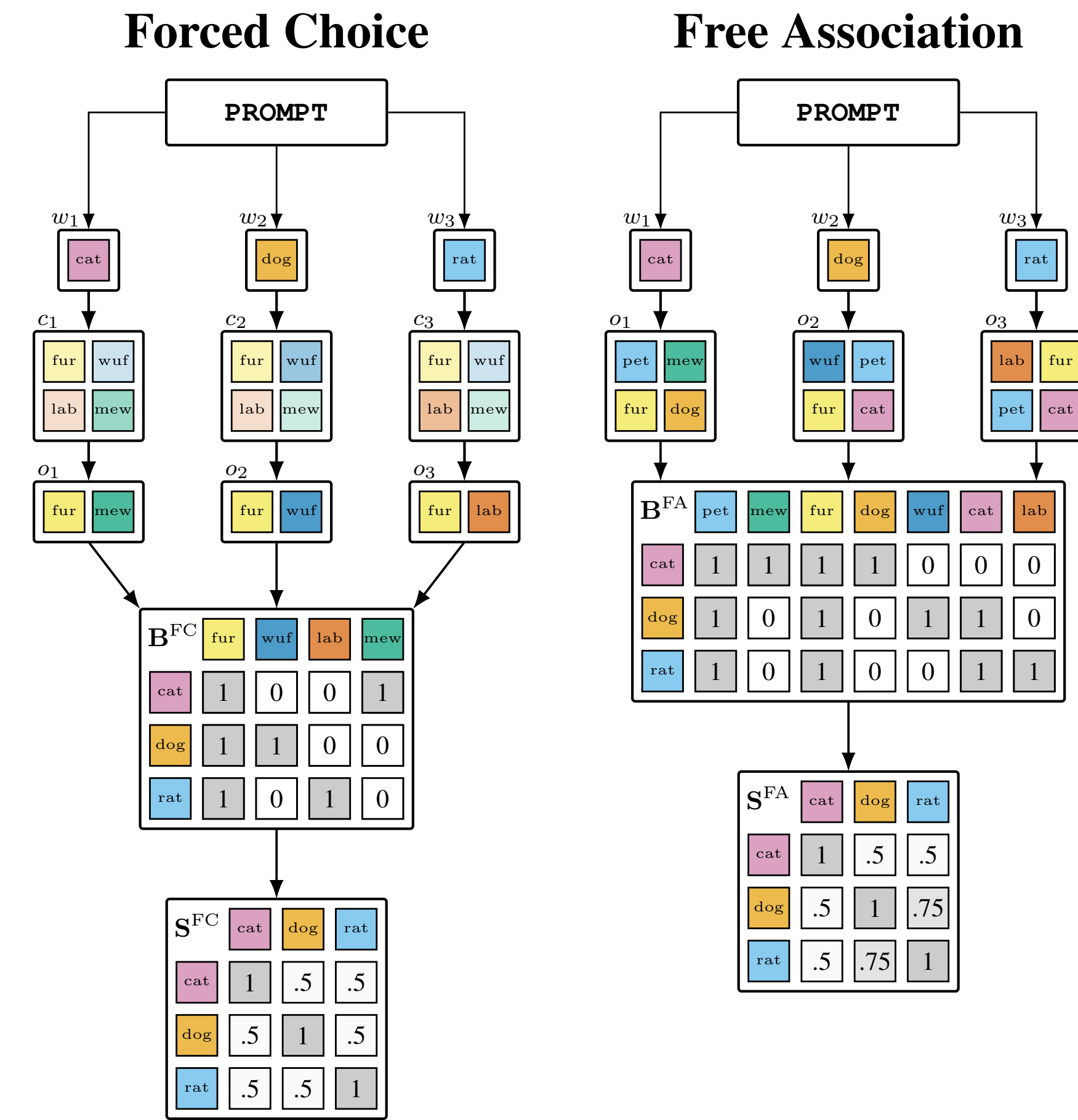


Figure 3. Forced Choice (FC): pick the 2 most related of 16 candidates $\rightarrow B^{\text{FC}}$. Free Association (FA): generate 5 associates $\rightarrow B^{\text{FA}}$. Cosine yields S^{FC} and S^{FA} .

Hidden-State Semantic Geometry

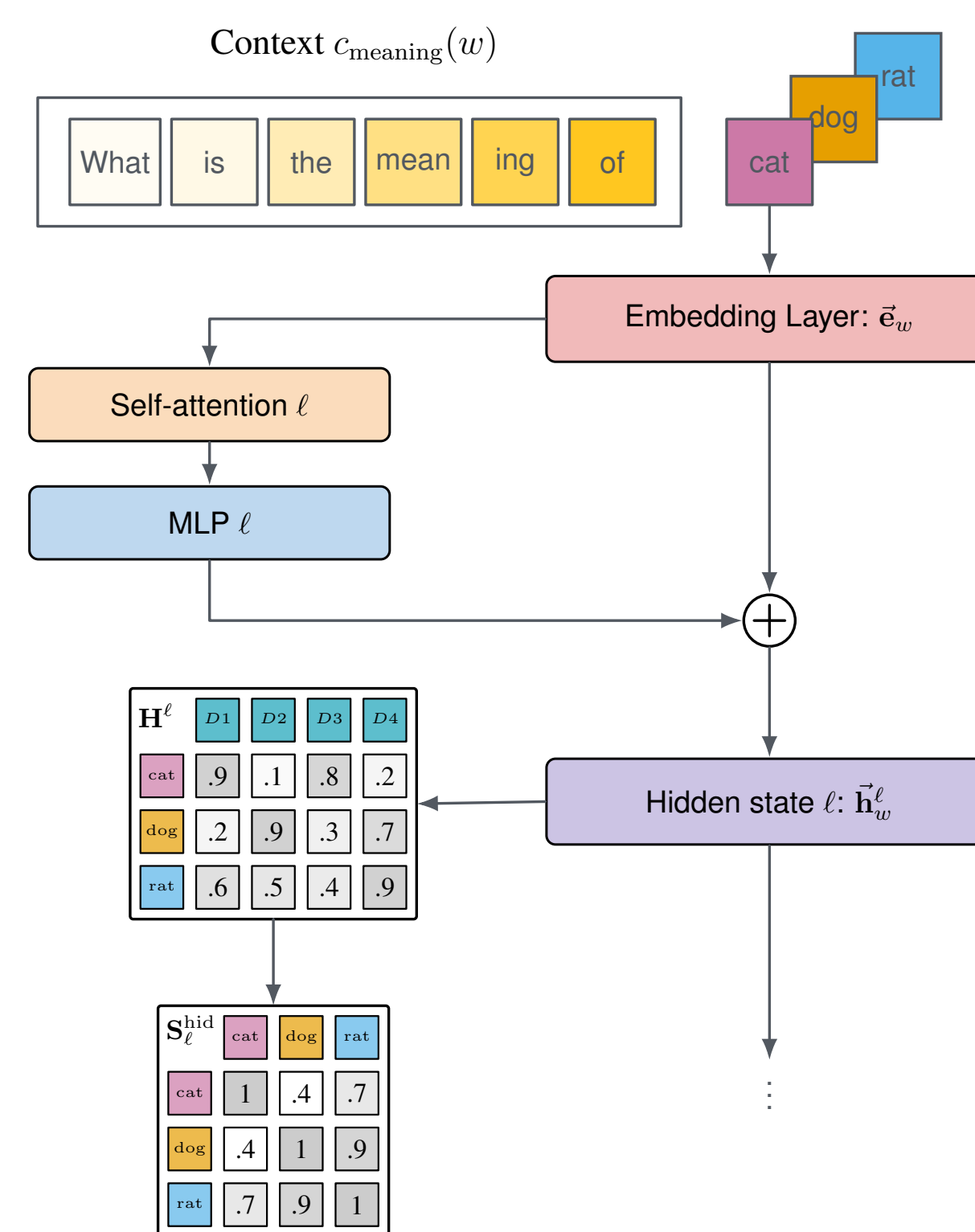


Figure 4. The *Meaning* prompt (“What is the meaning of w ?”): For $w \in \mathcal{V}$, extract $\bar{h}_w^{\ell} \rightarrow$ form matrix $H^{\ell} \rightarrow$ Cosine yields hidden-state similarity S_{ℓ}^{hid} .

Scale of the Study

- **Models:** 8 instruction-tuned LLMs.
- **Vocabulary:** 5,000 SUBTLEX-US nouns.
- **Trials:** 17.5M+ behavioral trials.
- **Hidden-state extraction:** 4 strategies for word w :
 - Meaning: “What is the meaning of w ?” (Figure 4).
 - Averaged: w in 50 natural sentences (C4).
 - Task (FC): w in the FC instruction prompt.
 - Task (FA): w in the FA instruction prompt.
- **Baselines:** FastText, BERT, Cross-Model Consensus.

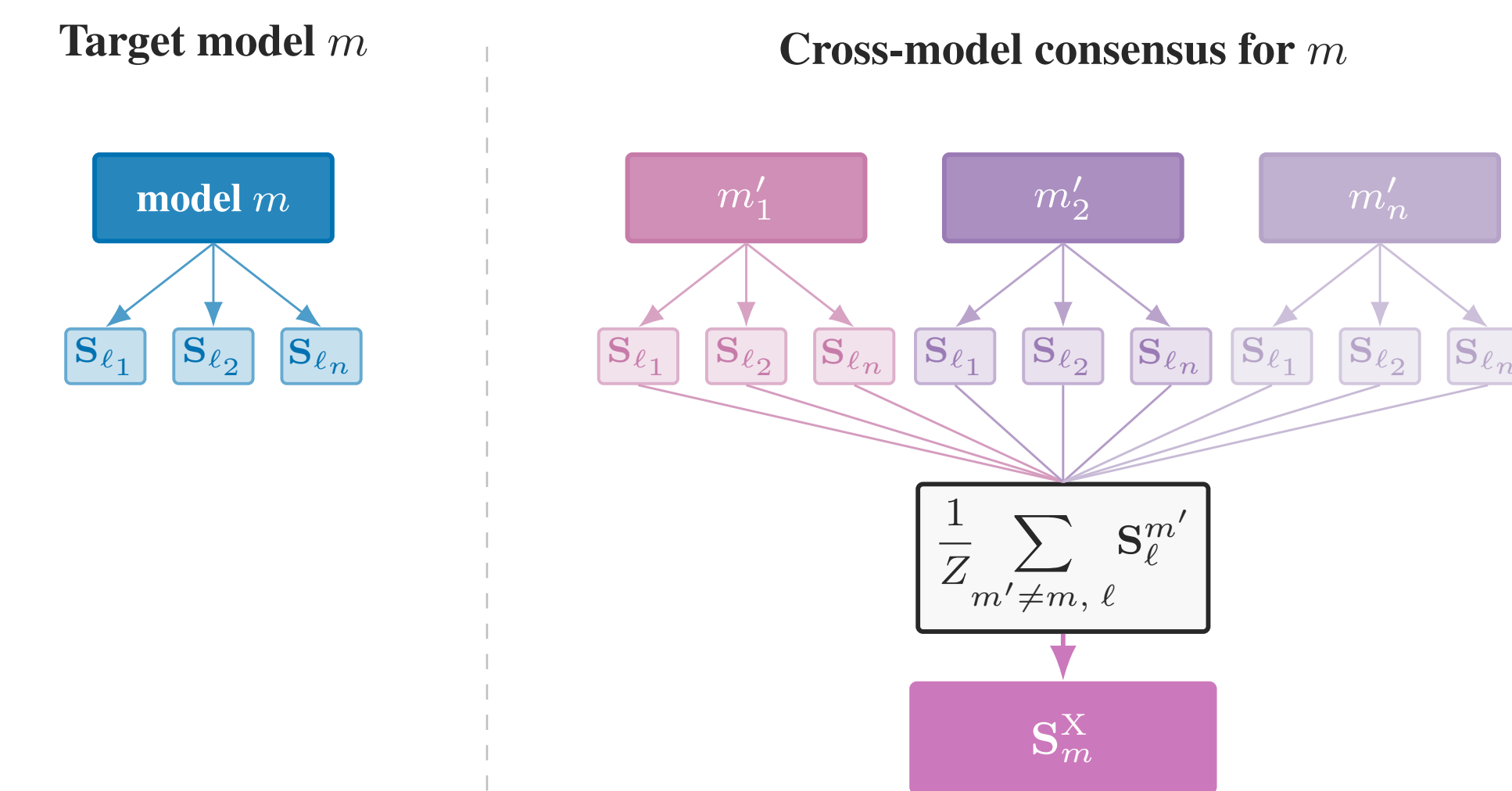


Figure 5. Cross-Model Consensus (CMC) averages the internal similarities of all *other* LLMs capturing a shared ‘platonic’ semantic geometry (Huh et al., 2024).

Result 1: Global & Local Alignment

FC aligns better with internal geometry than FA.

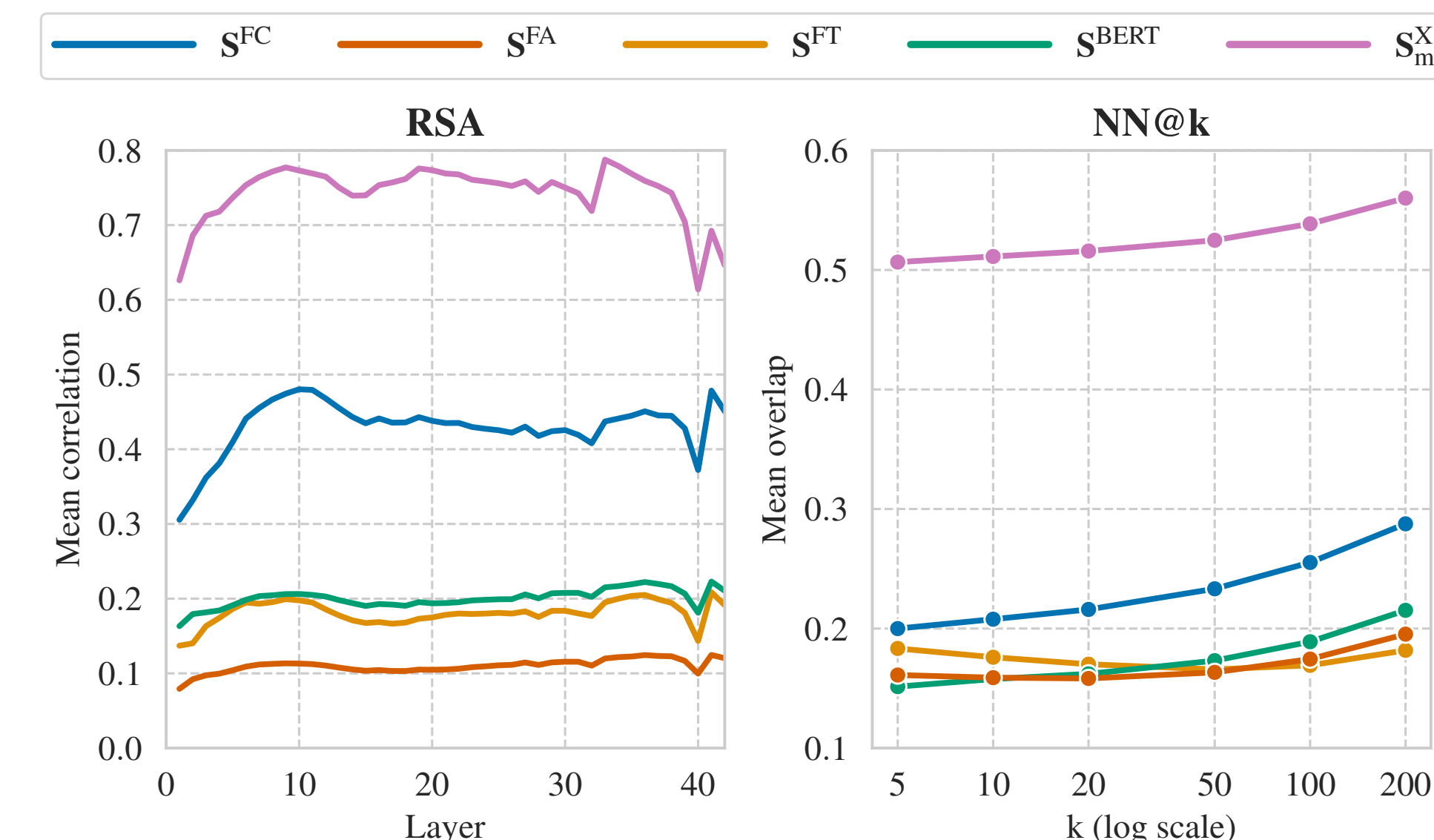


Figure 6. Layerwise RSA (left) and nearest-neighbor overlap $NN@k$ (right) over 8 models. Mean RSA: CMC .74, FC .43, FastText .20, FA .18, BERT .11.

Result 2: Predicting Unseen Words

FC predicts internal similarity beyond baselines.

	FC+FA	FC	FA
gemma-2-9b-it	.061	.059	.006
Llama-3.1-8B-Instruct	.033	.032	.003
rnj-1-instruct	.026	.026	.003
Mistral-7B-Instruct-v0.2	.025	.023	.004
Falcon3-10B-Instruct	.017	.017	.001
Qwen2.5-7B-Instruct	.009	.009	.000
Mistral-Nemo-Instruct-2407	.006	.006	.000
phi-4	.001	.001	.000
Mean	.022	.022	.002

Figure 7. Held-out words ridge ΔR^2 for behavioral measures beyond the FastText + BERT + CMC baselines.

Takeaways

- Behavior tracks internal geometry.** A model's own discrete text outputs read out a meaningful portion of its hidden-state geometry.
- Forced Choice > Free Association.** FC outperforms FA in all measures. Matrix structure is the lever: FC's dense matrix has a high signal-to-noise ratio; FA's sparse one does not.
- Consensus leads, behavior adds.** CMC aligns best of all with hidden-state similarities. FC still predicts internal similarities beyond baselines.
- No logits, activations, or weights required.** Behavioral similarity measures can be derived from closed/API-only models without access to internals.